Quiz-2: Foundations of RF and Microwave Sensors (S26.EC2.206)

Max. Marks: 40 **Time: 1 Hour** **Date: 31/03/2026**

NOTE: No queries are allowed. Write your assumptions (if any) for each question.

Q1 Sensors Basics (6 Marks)

(a) State the frequency range of an RF sensor. (1 Mark) — (300 MHZ — 300 GHZ)

(b) What is the input stimulus for an RF sensor.(1 Mark) — EM wave / M/E

(c) Define the following with one example each.(2 Marks)

(i) Sensor

(ii) Actuator

(d) Mention any one application of RF sensors. (1 Mark)

(e) State any one advantage of RF sensors over conventional sensors. (1 Mark)

Q2. Electromagnetic Wave Analysis (10 marks)

wave - speed

A uniform plane electromagnetic wave in free space is given by:

$$E(z,t) = 60\cos(4 \times 10^7 z - 1.2 \times 10^{10} t + \pi/6)\hat{y}$$

$v = 3 \times 10^8$ m/s

(a) Write the complex amplitude of the given wave. (1 Mark)

(b) Write the complete complex (phasor) representation of the wave. (2 Marks)

$v = \frac{\omega}{K} = 3 \ldots$ m

(c) From the given equation: (5 Marks)

Identify the amplitude and angular frequency (ω)

Calculate the following:

Wavelength (λ), Time period (T), Wave speed (v) give marks Both calc

(d)Write the expression for the corresponding magnetic field B(z,t) (2 Marks)

$B(z,t) = \frac{E_0}{v}$

$B(z,t) = 0.2 \cos(\ldots$

Q3 Attempt any TWO of the following (2 × 7 = 14 Marks)

(a)Using Maxwell's equations, derive the wave equation for the electric field in a dielectric medium. Clearly show all necessary steps. Also, write the corresponding wave equation for the magnetic field (no derivation required). (6 + 1 Marks)

(b) Define displacement current and derive its mathematical expression starting from Ampere's law. Further, explain its physical significance in electromagnetic wave propagation. (6 + 1 Marks)

(c) Starting from the integral form of Gauss's law, derive its differential form in terms of the electric displacement vector D. Also, explain the physical significance of a bound charge in a dielectric material. (6 + 1 Marks)

Q 4 MCQ Section (10 × 1 = 10 Marks)

i. Which of the following fields is always solenoidal?

(a) Electric field $\vec{E}$ (b) Magnetic field $\vec{B}$

(c) Displacement field $\vec{D}$ (d) Potential field

ii. The direction of EM wave propagation is given by:

(a) E + B (b) E − B

(c) E × H (d) B × H

iii. A time-varying electric field is given by

$$E(t) = 5\sin(10^6 t)\,\hat{y}\ \mathrm{V/m}$$

The displacement current density J_d
(Given: $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m)
(a) $4.425 \times 10^{-5}\cos(10^6 t)$ A/m²
(b) $8.85 \times 10^{-6}\sin(10^6 t)$ A/m²
(c) $4.425 \times 10^{-6}\cos(10^6 t)$ A/m²
(d) $8.85 \times 10^{-5}\cos(10^6 t)$ A/m²

iv. If the permittivity of a medium increases, the speed of the EM wave:
(a) Increases (b) Decreases
(c) Remains constant (d) Becomes zero

v. The direction of propagation of the wave

$$E(z,t) = E_0\cos(5z - 10t + \pi/4)\hat{y}$$

is:

(a) +z
(b) −y
(c) x
(d) No propagation

vi. Cartesian to cylindrical conversion for φ is:

(a) $\phi = \tan^{-1}(y/x)$
(b) $\phi = \cos^{-1}(z/r)$
(c) $\phi = \tan^{-1}(x/y)$
(d) $\phi = x + y$

vii. Which remains unchanged in cylindrical to spherical conversion?

(a) r (b) ϕ (c) θ (d) ρ

viii Cartesian to spherical conversion for θ is:

(a) $\theta = \tan^{-1}(y/x)$
(b) $\theta = \cos^{-1}(z/r)$
(c) $\theta = \sin^{-1}(z/r)$
(d) $\theta = x/y$

ix. Which of the following is true for electromagnetic waves?

(a) They require a material medium to propagate
(b) They are longitudinal in nature
(c) They are transverse and can propagate in a vacuum
(d) They have zero velocity

X. The continuity equation ensures that:

(a) The electric field is constant
(b) The magnetic field is zero
(c) Charge cannot be created or destroyed
(d) Current is always zero